

Question Paper

Exam Date & Time: 01-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Software Engineering [NPT_4123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the principles of abstraction and decomposition to explain how software engineers manage cognitive limitations while designing and maintaining complex software systems. Briefly describe how each principle simplifies understanding and problem-solving, and draw a schematic diagram that represents hierarchical abstraction levels and modular decomposition in a generic software system. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Analyze the concept of computer systems engineering and compare it with software engineering. Identify and discuss examples of product development projects where systems engineering would be most appropriately applied. (5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) Apply your understanding of the Evolutionary Development model to explain how it supports accommodating changing user requirements during software development. Illustrate your explanation with a suitable schematic representation of the model. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Software engineering has evolved through several programming paradigms-from early machine programming to aspect-oriented and embedded client-server designs." Apply your understanding of these design paradigms to illustrate, with suitable examples, how the shift from Control-Flow-Based Design to Object-Oriented modularity and maintainability of complex embedded client-server applications (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Explain the concept of cohesion in software design. Analyze and differentiate between temporal cohesion and sequential cohesion with suitable examples to show how each impacts module design. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) How would you identify and explain the level of coupling in a given software design? Is it true that in a good design, the modules should have low coupling? Why? (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Design an Alumni-Student Mentorship Web Portal that facilitates mentorship, career guidance, and (5)

- A) networking between current students and alumni of an institution. The system should allow alumni to register, create and update their professional profiles, and specify areas of expertise or mentorship interests. Students should be able to browse alumni profiles, send mentorship or career advice requests, and schedule virtual or in-person mentorship sessions. The system should also enable both alumni and students to communicate securely and track the status of mentorship requests. Additionally, the portal should maintain records of sessions and allow administrators to monitor overall engagement.

Using your understanding of UML Use Case Diagrams, apply the concepts to construct a use case diagram for this application

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Consider a Health Monitoring System that tracks a patient's vital signs (heart rate, blood pressure, oxygen level) through wearable sensors and sends alerts to doctors in case of abnormalities. Apply your understanding of Use Case Modelling to identify the actors and major use cases, and draw a schematic Component Diagram representing the system interactions. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Consider the following C function named sort. (5)

A)

```
/* sort takes an integer array and sorts it in ascending order */
void sort(int a[], int n){
    int i,j;
    for(i=0;i<n-1;i
        for(j=i+1;j<n;j++)
            if(a[i]>a[j])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
    }
```

Identify all linearly independent paths in the given program. Analyze how cyclomatic complexity can be used as a metric to determine the number of linearly independent paths for any program, and explain the steps involved in computing it.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Design a black-box test suite for a function that checks whether a character string (of up to twenty-five characters in length) is a palindrome. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4123:Software Engineering

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10

4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	NPT_4123:Software Engineering

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	1	5(10%)
4	Analysis	9	45(90%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)